

On Final Projects

- What? Anything You like
- Written report (repository) + presentation
- Either *Programming* or *Proposal*

Paper Abstract Format

Picosecond amorphization of SiO₂ stishovite under tension

Masaaki Misawa,^{1,2} Emina Ryuo,² Kimiko Yoshida,³ Rajiv K. Kalia,¹ Aiichiro Nakano,^{1*}
Norimasa Nishiyama,⁴ Fuyuki Shimojo,² Priya Vashishta,¹ Fumihiro Wakai³

Sci. Adv. **3**, e1602339 ('17)



(1) Problem: (1a) It is extremely difficult to realize two conflicting properties — high hardness and toughness — in one material. Nano-polycrystalline stishovite, recently synthesized from Earth-abundant silica glass, proved to be a super-hard, ultra-tough material, which could provide sustainable supply of high-performance ceramics. **(1b) However, its toughening mechanism remains elusive.** **(2) Finding:** Our quantum molecular dynamics simulations show that stishovite amorphizes rapidly on the order of picosecond under tension in front of a crack tip. **We find a displacive amorphization mechanism** that only involves short-distance collective motions of atoms, thereby facilitating the rapid transformation. The two-step amorphization pathway involves an intermediate state akin to experimentally suggested “high-density glass polymorphs”, before eventually transforming to normal glass. The rapid amorphization can catch up with, screen, and self-heal a fast moving crack. **(3) So What? This new concept of fast amorphization toughening likely operates in other pressure-synthesized hard solids.**

Tell your own narrative of history & place your work within!

Paper Abstract Format (2)

Towards Dynamic Simulations of Materials on Quantum Computers

Lindsay Bassman *et al.*, *Phys. Rev. B* **101**, 184305 ('20)

(1) Problem: (1a) With the recent experimental realization of quantum supremacy on a very specific problem, search is now on for the use of quantum computers for nontrivial scientific applications. A highly anticipated application is as a universal simulator of quantum many-body systems, as was conjectured by Richard Feynman in the 1980s and later elaborated by Seth Lloyd. The last decade has witnessed the growing success of quantum computing for simulating *static* properties of quantum systems, *i.e.*, the ground state energy of small molecules. **(1b) However, it remains a challenge to simulate quantum many-body dynamics on current-to-near-future noisy intermediate-scale quantum (NISQ) computers.** **(2) Finding:** Here, we demonstrate successful simulation of nontrivial quantum dynamics on publicly available NISQ computers, namely, IBM's Q16 Melbourne quantum processor and Rigetti's Aspen quantum processor. The compelling scientific problem is ultrafast control of emergent magnetism by THz radiation in an atomically-thin two-dimensional material. **(3) So What?** To liberate these newly available NISQ computers for broader scientific use, we also provide the full code and step-by-step tutorials for performing such simulations on each quantum processor. As such, **this work lays a foundation for the promising study of a wide variety of quantum dynamics on near-future quantum computers**, including dynamic localization of Floquet states and topological protection of qubits in noisy environments.

Problem funnel: Narrow down to the specific problem you solved!



Punch-Kick Writing

articulate_

WHAT WE DO +

HOW WE WORK

ABOUT US

BLOG

RESOURCES

CONTACT US

So what = kicker

HOW TO WRITE

Want to write well?
Open with a punch,
close with a kick

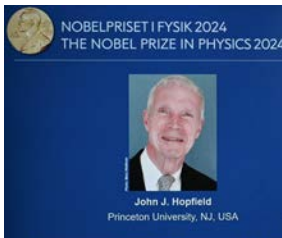


There are two words that every writer needs to know: lede and kicker. A 'lede' is the punchy opening sentence of an article. A 'kicker' is the last. If you can get them right, you can lift your writing to a whole new level.

<https://www.articulatemarketing.com/blog>

Winning Pattern in Science: Story

Why → So what → Now what



"Now what" by John Hopfield (*Princeton*)

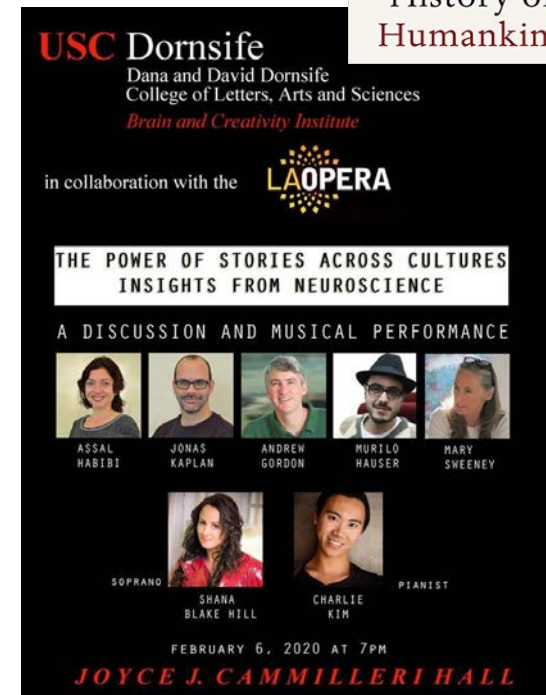
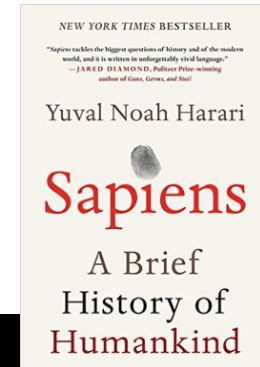
It's all telling a story!

"I concluded some time ago that the distinguishing characteristic of human intelligence is our story competence. We tell stories, we listen to stories, and we make up new stories by blending old ones together. That's really what education is all about, if you think about it...I think sharing the stories, the opinions, the asides, and understanding how a person solved a particular problem, what they were thinking of when they did that, what they were motivated by, etc. is just as, and probably more, important than teaching the actual skills."

"Open with a promise, close with a joke"
by Patrick Winston (*MIT*)

<http://www.ocw-openmatters.org/2016/07/19/open-with-a-promise-close-with-a-joke/>

Cognitive
revolution: ~70,000
BCE, Sapiens
evolved
imagination &
fictive language



Proposal ~ Qual: Neno's Ten Questions

1. What is the main goal of your work?
2. What are the tangible benefits?
3. What are the technical problems that make that goal difficult to achieve? (*i.e.*, why hasn't this been done already?)
4. What are the main elements of your approach?
5. How does your approach handle the technical problems that have prevented progress in the past? (*i.e.*, what makes you think you can do it when no one else could before?)
6. What are the unique, novel, and/or critical technologies developed in your approach?
7. What are the potential spin-offs or other applications of your work?
8. How can progress be measured? (*i.e.*, how can anyone tell if/when you've succeeded?)
9. What have you accomplished thus far?
10. What is your schedule for the work remaining?

“Answer all before you shall be allowed to take a qualifying exam.”

Prof. Nenad Medviovic (USC)



Manage Your Research with Paper

Whitesides' Group: Writing a Paper**

By George M. Whitesides*

Adv. Mater. 16, 1375 ('04)

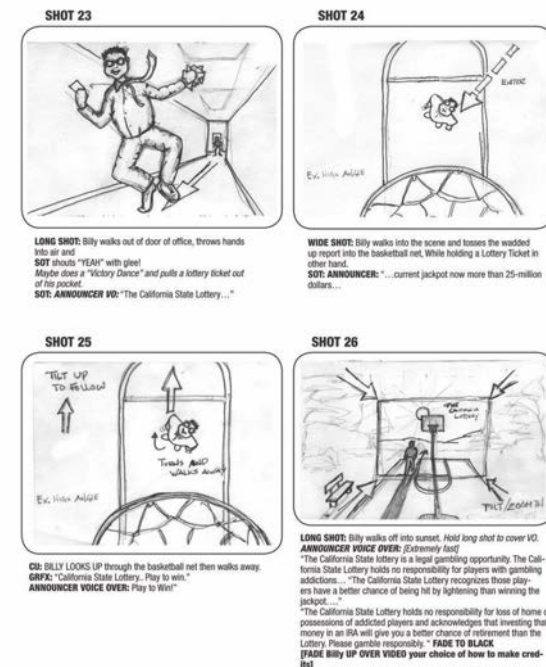
1. What is a Scientific Paper?

<https://aiichironakano.github.io/cs596/Whitesides-WritingPaper-AdvMater04.pdf>

A paper is an organized description of hypotheses, data and conclusions, intended to instruct the reader. Papers are a central part of research. If your research does not generate papers, it might just as well not have been done. “Interesting and unpublished” is equivalent to “non-existent”.

Realize that your objective in research is to formulate and test hypotheses, to draw conclusions from these tests, and to teach these conclusions to others. Your objective is not to “collect data”.

A paper is not just an archival device for storing a completed research program; it is also a structure for planning your research in progress. If you clearly understand the purpose and form of a paper, it can be immensely useful to you in *organizing* and conducting your research. A good outline for the paper is also a good plan for the research program. You should write and rewrite these plans/outlines throughout the course of the research. At the beginning, you will have mostly plan; at the end, mostly outline. The continuous effort to understand, analyze, summarize, and reformulate hypotheses on paper will be immensely more efficient for you than a process in which you collect data and only start to organize them when their collection is “complete”.



cf. Cinema storyboard

Do It for Your Final!

Team Project

- **Who did what (cf. SPMD)?** Team efforts are encouraged with the condition that the role of each team member is clearly delineated in the final-project report

Nature Geoscience **2**, 62 - 66 (2009)

Published online: 7 December 2008 | doi:10.1038/ngeo383

Subject Category: Geochemistry

Biomolecule formation by oceanic impacts on early Earth

Yoshihiro Furukawa¹, Toshimori Sekine², Masahiro Oba³, Takeshi Kakegawa¹ & Hiromoto Nakazawa²

Author contributions

H.N. proposed the impact synthesis hypothesis and conducted this study. Y.F. and T.S. carried out the shock recovery experiments. Y.F. extracted organic compounds and analysed amines and amino acids using LC–MS. M.O. and Y.F. analysed carboxylic acids using GC–MS. Y.F. and H.N. prepared an earlier manuscript. All authors discussed and prepared the final manuscript.

A CSCI596 Team

International Journal of Computational Science

1992-6669 (Print) 1992-6677 (Online) © Global Information Publisher

2007, Vol. 1, No. 4, 407-421

In situ visualization of parallel molecular dynamics

ParaViz: A Spatially Decomposed Parallel Visualization Algorithm Using Hierarchical Visibility Ordering

Cheng Zhang¹, Scott Callaghan², Thomas Jordan², Rajiv K. Kalia¹,

Aiichiro Nakano^{1*}, Priya Vashishta¹

material science

molecular dynamics

parallel

computer science

visualization

parallel

Who did what

Another CSCI596 Team

Computer Physics Communications 207 (2016) 186–192



Contents lists available at [ScienceDirect](#)

Computer Physics Communications

journal homepage: www.elsevier.com/locate/cpc



Parallel implementation of geometrical shock dynamics for two dimensional converging shock waves



Shi Qiu, Kuang Liu, Veronica Eliasson*

Aerospace and Mechanical Engineering, University of Southern California, Los Angeles, CA 90089-1191, USA

mechanical engineering

computer science

shock physics Who did what parallel

[Computer Physics Communications homepage](#)

Multi-Class Project

APPLIED PHYSICS LETTERS **100**, 163108 (2012)

Critical dimensions of highly lattice mismatched semiconductor nanowires grown in strain-releasing configurations

Suzana Sburlan,¹ P. Daniel Dapkus,^{1,2} and Aiichiro Nakano^{2,3}

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³*Collaboratory for Advanced Computing and Simulations, Department of Computer Science, University of Southern California, Los Angeles, California 90089-0242, USA*

JOURNAL OF APPLIED PHYSICS **111**, 054907 (2012)

Effect of substrate strain on critical dimensions of highly lattice mismatched defect-free nanorods

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***O(N)* Lanczos eigensolver (PHYS516) → parallelization (CSCI596/653)**

Multi-Class Project (2)

Computer Physics Communications 219 (2017) 246–254



Contents lists available at ScienceDirect

Computer Physics Communications

journal homepage: www.elsevier.com/locate/cpc



A derivation and scalable implementation of the synchronous parallel kinetic Monte Carlo method for simulating long-time dynamics



Hye Suk Byun^a, Mohamed Y. El-Naggar^{a,b,c}, Rajiv K. Kalia^{a,d,e,f}, Aiichiro Nakano^{a,b,d,e,f,*},
Priya Vashishta^{a,d,e,f}

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^f Collaboratory for Advanced Computing and Simulations, University of Southern California, Los Angeles, CA 90089-0242, USA

Kinetic Monte Carlo (PHYS516) → parallelization (CSCI596/653)

More Class Projects



PAR²: Parallel Random Walk Particle Tracking Method for solute transport in porous media[☆]

Calogero B. Rizzo^{a,*}, Aiichiro Nakano^b, Felipe P.J. de Barros^a



Buildings **9**, 44 (2019)



Article

Adaptive Kinetic Architecture and Collective Behavior: A Dynamic Analysis for Emergency Evacuation

Angella Johnson^{1,*}, Size Zheng², Aiichiro Nakano³, Goetz Schierle¹ and Joon-Ho Choi¹



WaterAlignment: Identification of displaced water molecules in molecular docking using Jonker and Volgenant shortest path augmentation for linear assignment[☆]

Dab Brill^{a,*}, Jason B. Giles^c, Ian S. Haworth^c, Aiichiro Nakano^b



Computer Physics Communications **247** (2020) 106873

sDMD: An open source program for discontinuous molecular dynamics simulation of protein folding and aggregation[☆]

Size Zheng^{a,*}, Leili Javidpour^b, Muhammad Sahimi^c, Katherine S. Shing^c, Aiichiro Nakano^c



Boltzmann machine modeling of layered MoS₂ synthesis on a quantum annealer

Jeremy Liu^{a,b}, Ankith Mohan^b, Rajiv K. Kalia^c, Aiichiro Nakano^c, Ken-ichi Nomura^{c,*}, Priya Vashishta^c, Ke-Thia Yao^d



Geophysical Research Letters **48**, e2020GL091681 (2021)

Molecular Dynamics Simulations of Dielectric Breakdown of Lunar Regolith: Implications for Water Ice Formation on Lunar Surface

Ziyu Huang¹, Ken-ichi Nomura², Aiichiro Nakano³, and Joseph Wang¹

More Class Projects (2)

Article

<https://doi.org/10.1038/s41467-024-47685-8>

Scalable computation of anisotropic vibrations for large macromolecular assemblies

Jordy Homing Lam ^{1,2,3}, Aiichiro Nakano ^{1,4,5}  & Vsevolod Katritch ^{1,2,3,6} 

Nature Communications | (2024)15:3479

JCIM
JOURNAL OF
CHEMICAL INFORMATION
AND MODELING

COMPUTATIONAL CHEMISTRY | September 11, 2024

pubs.acs.org/jcim

Article

Exploring the Global Reaction Coordinate for Retinal Photoisomerization: A Graph Theory-Based Machine Learning Approach

Goran Giudetti,[⊥] Madhubani Mukherjee,[⊥] Samprita Nandi, Sraddha Agrawal, Oleg V. Prezhdo, and Aiichiro Nakano*



Cite This: <https://doi.org/10.1021/acs.jcim.4c00325>

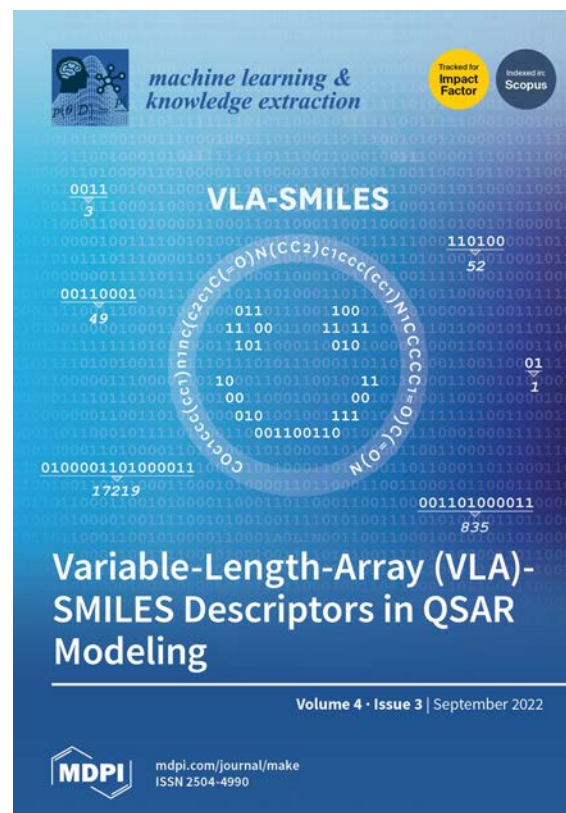


Read Online

More Class Projects: Journal Cover



Nazarova *et al.*,
JCIIM **61**, 2175 ('21)



Nazarova *et al.*,
MLKE **4**, 715 ('22)

Project in Another Class

ARTICLE OPEN

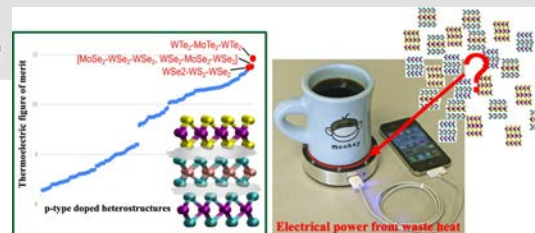
Active learning for accelerated design of layered materials

Lindsay Bassman^{1,2}, Pankaj Rajak^{1,3}, Rajiv K. Kalia^{1,2,3,4}, Aiichiro Nakano^{1,2,3,4,5}, Fei Sha^{4,5}, Jifeng Sun⁶, David J. Singh⁶, Muratahan Aykol⁷, Patrick Huck⁷, Kristin Persson⁷ and Priya Vashishta^{1,2,3,4}

Hetero-structures made from vertically stacked monolayers of transition metal dichalcogenides hold great potential for optoelectronic and thermoelectric devices. Discovery of the optimal layered material for specific applications necessitates the estimation of key material properties, such as electronic band structure and thermal transport coefficients. However, screening of material properties via brute force ab initio calculations of the entire material structure space exceeds the limits of current computing resources. Moreover, the functional dependence of material properties on the structures is often complicated, making simplistic statistical procedures for prediction difficult to employ without large amounts of data collection. Here, we present a Gaussian process regression model, which predicts material properties of an input hetero-structure, as well as an active learning model based on Bayesian optimization, which can efficiently discover the optimal hetero-structure using a minimal number of ab initio calculations. The electronic band gap, conduction/valence band dispersions, and thermoelectric performance are used as representative material properties for prediction and optimization. The Materials Project platform is used for electronic structure computation, while the BoltzTraP code is used to compute thermoelectric properties. Bayesian optimization is shown to significantly reduce the computational cost of discovering the optimal structure when compared with finding an optimal structure by building a regression model to predict material properties. The models can be used for predictions with respect to any material property and our software, including data preparation code based on the Python Materials Genomics (PyMatGen) library as well as python-based machine learning code, is available open source.

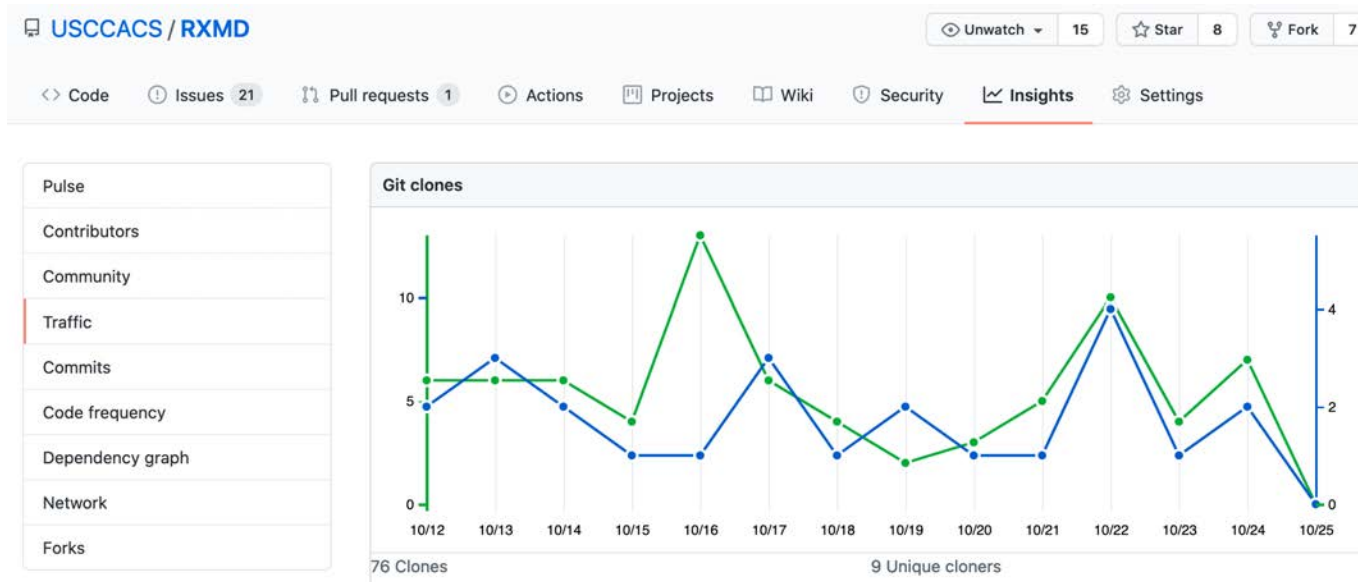
npj Computational Materials (2018)4:74; <https://doi.org/10.1038/s41524-018-0129-0>

Involve your Ph.D. advisor in final discussion!



Final Project FAQ

- Q.** How to make my GitHub repository look good to recruiters?
- A.** Make a lot of commits to making your software active, rather than just a one-time dead software
- Q.** How can I see my GitHub repository is active?
- A.** Go to “insights” of your repository & choose “contributors” (who do works), “code frequency” or “traffic” (how many are accessing)



Review GitHub lecture: <https://aiichironakano.github.io/cs596/Git-VG.pdf>